SOFTWARE ENGINEERING LAB-1

SRN: PES1UG25CS847

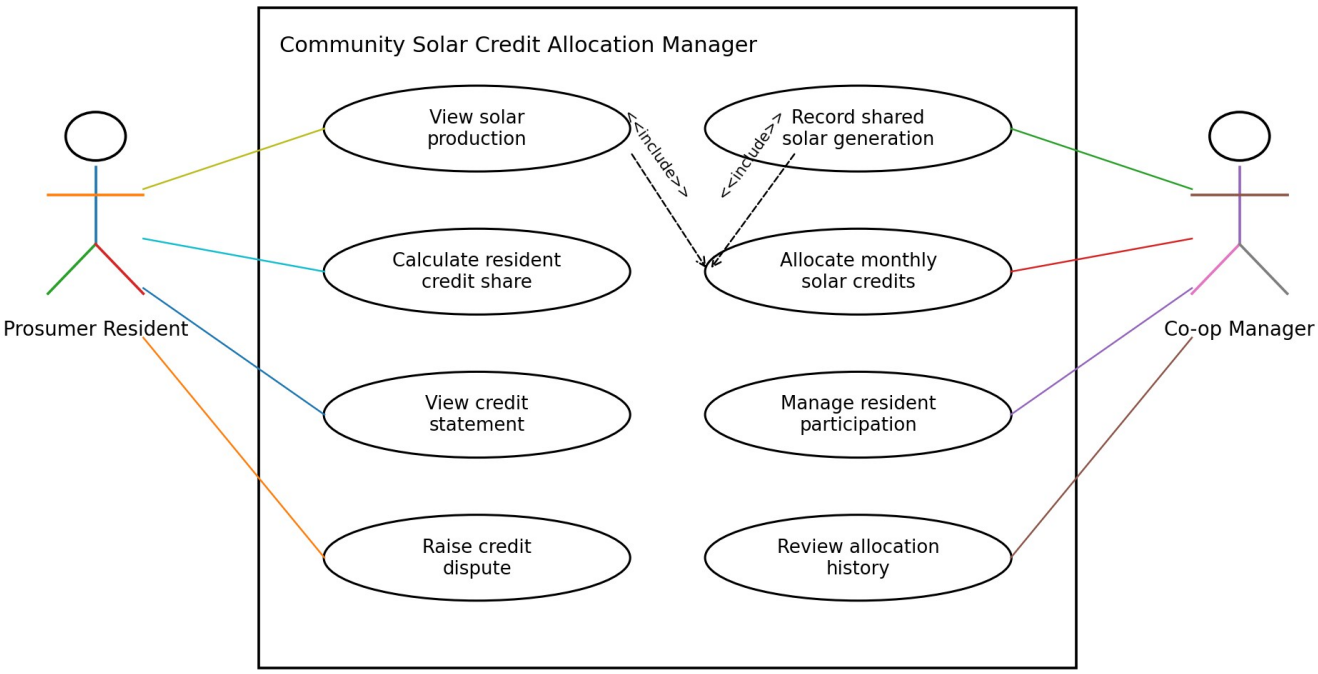
Project: Community Solar Credit Allocation Manager

Domain: Sustainability & Green Tech

1. Requirements Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall register participating prosumer residents and their solar-share details.	High	A resident can be added with a unique ID and valid participation/share data.	Required to identify who is eligible to receive community-solar credits.
FR-002	Functional	The system shall record the total energy generated by the shared solar installation for each billing period.	High	The co-op manager can enter/import generation in kWh for a selected month.	Monthly generation is the basis for credit allocation.
FR-003	Functional	The system shall calculate each resident's credit using the approved allocation rule and participation share.	High	For a billing period, the calculated resident credits sum to the distributable community credit.	Ensures fair and consistent distribution among participants.
FR-004	Functional	The system shall allocate the calculated solar credit to each eligible resident for the billing period.	High	Each eligible resident receives one traceable credit allocation for the selected month.	Provides the main business function of the system.
FR-005	Functional	Residents shall be able to view monthly generation, allocated credits, and allocation history.	Medium	A resident can view current and previous credit statements with period and credit value.	Improves transparency for prosumers.
FR-006	Functional	The co-op manager shall manage resident participation status and allocation parameters.	Medium	Manager can activate/deactivate residents and update valid share data before allocation.	Keeps allocation inputs accurate when membership changes.
FR-007	Functional	Residents shall be able to raise a dispute for an allocation and the manager shall record its resolution.	Medium	A dispute is linked to a billing period and has Open/Resolved status with remarks.	Supports correction and accountability for disputed credits.
NFR-001	Non-Functional - Performance	The system shall generate a monthly allocation for all registered residents within 5 seconds for normal lab-scale data.	Medium	A standard allocation run completes within 5 seconds.	Keeps the system responsive during monthly processing.
NFR-002	Non-Functional - Reliability/Security	Allocation records shall be access-controlled and retained with an audit trail.	High	Residents see only their records; manager actions and allocation changes are logged.	Protects resident data and preserves trust in credit calculations.

2. UML Use Case Diagram



3. Use-Case Flow Specification

Use Case: Monthly Community Solar Credit Allocation

Actors: Prosumer Resident, Co-op Manager

1. Preconditions

The community solar installation and the billing period are configured.

Participating prosumer residents and their valid allocation shares are registered.

The co-op manager has the generation data for the selected billing period.

2. Postconditions

Solar credits are calculated and allocated to all eligible prosumer residents.

Each allocation is stored with the billing period, generation data, resident share, and calculated credit.

Residents can view the resulting credit statement and allocation history.

3. Main Success Scenario

1. The co-op manager selects the billing period.
2. The system retrieves the active participating residents and their allocation shares.
3. The co-op manager records or confirms the total shared solar generation for the period.
4. The system validates the generation data and participation information.
5. The system calculates each resident's proportional solar credit according to the approved allocation rule.
6. The system displays the proposed allocation summary to the co-op manager.
7. The co-op manager confirms the allocation.
8. The system stores the allocation records and makes each resident's credit statement available.
9. The prosumer resident views the monthly credit and allocation history.

4. Alternate Flow

A1. Invalid or Missing Generation Data: If generation data is missing or invalid, the system stops allocation and asks the co-op manager to correct the input.

A2. Resident Participation Changed: If a resident is inactive or has an invalid share for the selected period, the system excludes/flags that resident until the participation data is corrected.

A3. Credit Dispute: A prosumer resident may raise a dispute against a monthly allocation. The system records the dispute and the co-op manager reviews it, records remarks, and marks it resolved after correction or confirmation.

Basis

The system supports transparent and fair distribution of community-solar benefits by converting shared generation into traceable credits for participating residents, while allowing the co-op manager to maintain participation data and resolve allocation issues.

GitRepo Link: Add your repository link here after uploading the Lab-1 files.